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<!DOCTYPE html>
<html lang="ko">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0,
maximum-scale=1.0, user-scalable=no">
  <title>Arrow Trajectory Simulator</title>
  <link rel="stylesheet" href="style.css">
  <link rel="manifest" href="manifest.json">
  <link rel="icon" href="icon-192.png" type="image/png">
<script>
  if ('serviceWorker' in navigator) {
    navigator.serviceWorker.register('sw.js')
      .then(() => console.log("서비스 워커 등록 완료"))
      .catch(err => console.error("서비스 워커 등록 실패", err));
  }
</script>
</head>
<body>
<!-- 인트로 및 공지사항 모달 팝업 -->
<div id="introModal" class="intro-modal">
  <div class="intro-content">
    <div class="intro-header">
      <span class="intro-logo">🚀</span>
      <h2>국궁 탄도 시뮬레이터</h2>
    </div>
    <div class="intro-body">
      <h3>📄 시스템 공지사항</h3>
      <ul>
        <li><strong>버그 수정 완료:</strong> 측면 궤적과 과녁 확대도 판정이 일치하지
않던 기하학 오류를 수정했습니다.</li>
        <li><strong>환경 설정 안내:</strong> 바람, 과녁 고도차, 사수의 위치를 변경하
며 정밀한 조준점 변화를 테스트해 보세요.</li>
        <li>본 프로그램은 국궁 표준 제원을 바탕으로 제작된 시뮬레이터입니다.</li>
      </ul>
    </div>
    <div class="intro-footer">
      <button class="intro-close-btn" onclick="closeIntro()">시뮬레이터 시작하기
</button>
    </div>
  </div>
</div>
</div>

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<!-- 상단 60% 영역: 메인 스크린 및 시뮬레이터 캔버스 -->

<div class="main-screen">

<h1>Arrow Trajectory</h1>

<div class="segmented-control">

<button class="segment-btn" onclick="changeView('front', this)">정면</button>

<button class="segment-btn active" onclick="changeView('side', this)">측면

</button>

<button class="segment-btn" onclick="changeView('top', this)">평면</button>

<button class="segment-btn" onclick="changeView('target', this)">과녁

</button>

</div>

<div class="sim-container">

<canvas id="simCanvas"></canvas>

<button id="draggableFireBtn" class="fire-btn">발사</button>

</div>

</div>

<!-- 하단 40% 영역: 라운드 카드형 설정 및 결과 패널 컨테이너 -->

<div id="panel-container">

<!-- [탭 1] 화살 설정 -->

<div class="fixed-panel active" id="panel-arrow">

<div class="panel-card">

<div class="panel-title">화살 설정</div>

<div class="form-grid">

<div class="form-group"><label>화살 무게</label><div class="input-wrapper"><input type="number" id="weight" value="25">g</div></div>

<div class="form-group"><label>화살 직경</label><div class="input-wrapper"><input type="number" id="diameter" value="5.5" step="0.1">mm</div></div>

<div class="form-group"><label>항력 계수</label><div class="input-wrapper"><input type="number" id="dragCoeff" value="0.35" step="0.01">cd</div></div>

<div class="form-group"><label>양력 계수</label><div class="input-wrapper"><input type="number" id="liftCoeff" value="0.05" step="0.01">cl</div></div>

</div>

</div>

</div>

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<!-- [탭 2] 사법 설정 -->
<div class="fixed-panel" id="panel-method">
  <div class="panel-card">
    <div class="panel-title">사법 설정</div>
    <div class="form-grid">
      <div class="form-group">
        <label>발사 각도</label>
        <div class="input-wrapper">
          <input type="text" id="angle" value="30" inputmode="text"
pattern="^-?[0-9]*\.[0-9]*$" >
            <span class="unit">°</span>
          </div>
        </div>
        <div class="form-group"><label>초기 속도</label> <div
class="input-wrapper"><input type="number" id="velocity" value="50"> <span
class="unit">m/s</span> </div> </div>
        <div class="form-group">
          <label>수평 각도</label>
          <div class="input-wrapper">
            <input type="text" id="yawAngle" value="0" inputmode="text"
pattern="^-?[0-9]*\.[0-9]*$" >
              <span class="unit">°</span>
            </div>
          </div>
        <div class="form-group">
          <label>발사 높이 </label>
          <div class="input-wrapper">
            <input type="number" id="launchHeight" value="1.5" step="0.1">
            <span class="unit">m</span>
          </div>
        </div>
        <div class="form-group">
          <label for="launchZ">사수 위치 </label>
          <div class="input-wrapper">
            <input type="text" id="launchZ" value="0" min="-10" max="10"
step="0.5">
              <span class="unit">m</span>
            </div>
          </div>
        </div>
      </div>
    </div>
  </div>
</div>

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<!-- [탭 3] 환경 설정 -->
<div class="fixed-panel" id="panel-env">
  <div class="panel-card">
    <div class="panel-title">환경 설정</div>
    <div class="form-grid">
      <div class="form-group">
        <label>종풍</label>
        <div class="input-wrapper">
          <input type="text" id="windX" value="0" inputmode="text"
pattern="^-?[0-9]*W?[0-9]*$" >
          <span class="unit">m/s</span>
        </div>
      </div>
      <div class="form-group">
        <label>횡풍</label>
        <div class="input-wrapper">
          <input type="text" id="windY" value="0" inputmode="text"
pattern="^-?[0-9]*W?[0-9]*$" >
          <span class="unit">m/s</span>
        </div>
      </div>
      <div class="form-group">
        <label>과녁 고도차</label>
        <div class="input-wrapper">
          <input type="text" id="targetHeight" value="1.3" inputmode="text"
pattern="^-?[0-9]*W?[0-9]*$" >
          <span class="unit">m</span>
        </div>
      </div>
      <div class="form-group"><label>공기 밀도</label><div
class="input-wrapper"><input type="number" id="airDensity" value="1.225"
step="0.001"><span class="unit">kg/m³</span></div></div>
    </div>
  </div>
</div>

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<!-- [탭 4] 비행 결과 -->

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<div class="fixed-panel" id="panel-result">
  <div class="panel-card">
    <div class="panel-title">비행 결과 분석</div>
    <div class="form-grid">
      <div class="form-group"><label>최대 거리</label><div

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class="result-value" id="resMaxDist">0.00 m</div> </div>
    <div class="form-group"><label>최대 높이</label> <div
class="result-value" id="resMaxHeight">0.00 m</div> </div>
    <div class="form-group"><label>측면 편차</label> <div
class="result-value" id="resSideDev">0.00 m</div> </div>
    <div class="form-group"><label>비행 시간</label> <div
class="result-value" id="resFlightTime">0.00 s</div> </div>
    <div class="form-group"><label>충돌 속도</label> <div
class="result-value" id="resImpactVel">0.00 m/s</div> </div>
    <div class="form-group"><label>충돌 에너지</label> <div
class="result-value" id="resImpactEnergy">0.00 J</div> </div>
    </div>
  </div>
</div>

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<!-- 📍 [위치 이동] 설정 패널 밖으로 완전히 탈출시켜 맨 아래 고정된 저작권 표시 영역
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<div id="app-copyright">
  Copyright © 2026 해현-활터왔소. All Rights Reserved.
</div>

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<!-- 맨 하단 메뉴 전환 탭바 -->
<div class="tab-bar">
  <button class="tab-item active" onclick="switchPanel('arrow')"><span
class="tab-icon">🏠</span> <span>화살 설정</span> </button>
  <button class="tab-item" onclick="switchPanel('method')"><span
class="tab-icon">🔧</span> <span>사법 설정</span> </button>
  <button class="tab-item" onclick="switchPanel('env')"><span
class="tab-icon">🌿</span> <span>환경 설정</span> </button>
  <button class="tab-item" onclick="switchPanel('result')"><span
class="tab-icon">📊</span> <span>비행 결과</span> </button>
</div>

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<script src="ui.js"></script>
<script src="physics.js"></script>
</body>
</html>

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